A Unified Deep Learning Network for Remote Sensing Image Registration and Change Detection

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***Abstract*— Image registration and change detection are crucial for multitemporal remote sensing image analysis. The images should be registered before the change information detection. Existing deep learning methods have shown significant advantages in image registration and change detection tasks. They usually design two independent task-specific deep networks for image registration and change detection, respectively. These independent deep networks will learn from scratch and rely on many task-specific labeled training datasets. This article finds that image registration and change detection have similar learning mechanisms, which focus on extracting discriminative features. Inspired by this, we propose a Unified image Registration and Change detection Network (URCNet) that can perform image alignment and change information detection through a single network. Additionally, this article proposes various deep collaborative learning methods for URCNet optimization, which enforce that the URCNet can effectively support remote sensing image registration and change detection simultaneously. Extensive experiments demonstrate the effectiveness of the proposed URCNet for image registration and change detection, which can achieve comparable and better results with task-specific and more complex deep networks. The proposed URCNet can support multitasks based on the same scene images, different scene images, and even multimodal images. Moreover, URCNet shows significant advantages over other deep networks in change detection under limited labeled datasets.**

***Index Terms*— Change detection, deep collaborative learning, image registration, unified deep network.**

I. Introduction

IMAGE registration and change detection are crucial for remote sensing image applications, such as land resource management and disaster information estimation [1], [2].

Manuscript received 4 September 2023; revised 23 November 2023; accepted 12 December 2023. Date of publication 19 December 2023; date of current version 3 January 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62201407, Grant 62271377, and Grant 62171347; in part by the National Key Research and Development Program of China under Grant 2021ZD0110400; in part by the Key Research and Development Program of Shannxi under Grant 2021ZDLGY0106 and Grant 2022ZDLGY0112; and in part by the Key Scientific Technological Innovation Research Project by Ministry of Education. *(Corresponding author: Dou Quan.)*

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Digital Object Identifier 10.1109/TGRS.2023.3344751

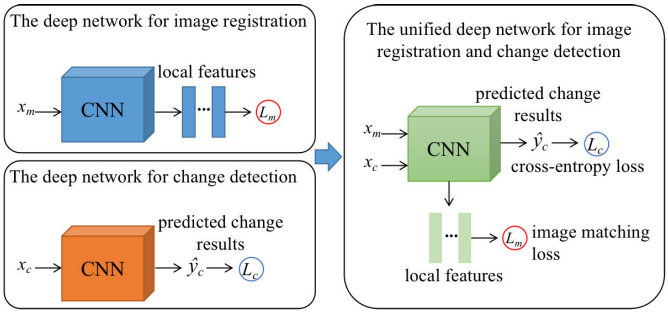


Fig. 1. Existing methods usually design two task-specific deep networks for image registration and change detection, respectively. This article proposes a unified deep network for these two tasks.

Image registration focuses on aligning images acquired at different times, from different viewpoints, or by various sensors [3], [4], which is widely used for multitemporal image analysis, multiview image processing, and multimodal image fusion. Image change detection task aims to identify the change information between the aligned multitemporal images [5], [6]. Thus, image registration is the basis of change detection.

Generally, researchers deal with image registration and change detection tasks independently and develop image registration methods and change detection methods, respectively. Image registration methods can be categorized into intensity-based methods and features-based methods. Intensity-based methods directly search the optimal registration matrix by comparing the image intensity information, according to the similarity metrics of mutual information (MI) [7] and cross correlation (CC) [8]. The intensity information is easily acquired but sensitive to illumination changes and noise. Feature-based methods mainly find the local correspondences between two images according to the similarity of features. The traditional representative features used in image registration include local texture, structure information, and statistics of gradients [9], [10], [11], [12], [13]. The features are more robust to image changes than image intensity and are widely used for remote sensing image registration. Image change detection methods mainly identify the change information between multitemporal images according to the image pixels or features. Traditional change detection methods adopt clustering or threshold segmentation techniques to generate the binary detection results for each pixel based on the difference maps. The remote sensing image will be classified into changed or unchanged regions.

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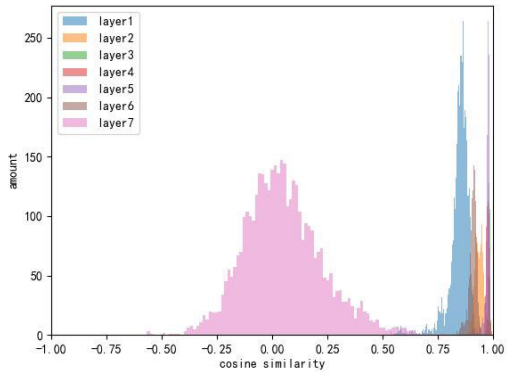


Fig. 2. Cosine similarity between multilayer deep features extracted by the image registration network and the change detection network.

Due to the deep convolutional network that can extract robust and discriminative features, deep learning methods have achieved significant advantages in image registration and change detection tasks. The deep convolutional network extracts local discriminative features for matching image patches and building local correspondences in the image registration task. It can be optimized by the matching loss that maximizes the feature distance of nonmatching and minimizes the feature distance of matching samples [14], [15]. In the change detection task, the deep convolutional network extracts discriminative features and identifies the change information according to the features of the multitemporal images. The change detection task can be viewed as a binary classification problem. Thus, we can use the cross-entropy loss to optimize the change detection network.

Existing deep learning methods design two fully independent task-specific networks for image registration and change detection, respectively. As shown in Fig. 1, these two independent deep networks have different network parameters without any information interaction in the training process. The optimization of task-specific deep networks relies on a lot of labeled training datasets. Their performances will significantly decrease under the limited labeled training datasets. Meanwhile, we usually adopt two separate deep networks for image registration and change detection in the inference phase. Thus, existing task-specific deep networks will increase the cost of parameter storage and training dataset labeling for remote sensing image applications. Actually, image registration and change detection have strong correlations and have the potential to be unified.

Inspired by this, this article proposes to build a unified deep framework for image registration and change detection tasks. First, deep learning methods for image registration and change detection focus on extracting discriminative features from images to compare the feature similarity or predict change labels. It is natural to unify these two tasks by sharing the feature extraction model. We extract the multilayer deep features by the trained deep network for image registration and change detection from 4000 pairs of remote sensing images and calculate their cosine similarity between these deep features, as shown in Fig. 2. It can be seen that most of the deep features have high cosine similarity, except for high-layer features. The deep features with high similarity can be viewed as task-independent features, while the high features with small similarity can be viewed as task-dependent features. The task-independent features can be shared for different remote sensing image processing tasks, enhancing deep network optimization. Thus, it is possible to build a unified deep network for remote sensing image registration and change detection.

However, it is difficult to build a unified network that can perform well for various remote sensing image processing tasks because they have different optimization goals. Additionally, image registration and change detection tasks may need to deal with different scene images and even multimodal images in practical applications, further increasing the challenges of building a unified network. For example, the unified network may perform the image registration on multimodal images, for example, optical (OPT) and synthetic aperture radar (SAR) images, and conduct the image change detection task on the single-modal images, for example, OPT remote sensing images. The complex unified network with more learnable parameters may perform well on remote sensing image processing tasks, while it will significantly increase the computation costs. Thus, it is challenging to build a unified network that is effective for image registration and change detection tasks with a small computation cost burden.

To achieve this, we construct a unified deep network for image registration and change detection, named URCNet, where these two tasks share the feature extraction block. For the image registration task, the URCNet can extract the local features to build the local matches and align images. For the change detection task, the URCNet learns the deep features from the multitemporal images and generates the corresponding change map. To enforce the URCNet to support the image registration and change detection tasks simultaneously, we adopt the matching loss and cross-entropy loss for URCNet training, as shown in Fig. 1 (right). Additionally, this article proposes different deep collaborative learning methods for URCNet optimization, including the frozen learning method, fine-tuning method, and continuous learning method. The URCNet optimized by the deep collaborative learning method is effective and robust for image registration and change detection tasks under various application scenarios, including multitask based on the same scene (e.g., OPT image registration and OPT image change detection) and multitask based on different scenes, even multimodal images (e.g., OPT-SAR image registration and OPT image change detection). The URCNet can acquire comparable and better performances with the more complex deep network for change detection. The proposed URCNet can also prevent significant performance degradation as the decrease of labeled datasets and achieve the best performances under limited labeled change detection datasets.

The main contributions can be summarized as follows.

1) This article first proposes a unified deep network, URCNet, for image registration and change detection tasks. Different from existing deep learning networks that are task-specific and merely focus on improving the performances on a single task of image registration or change detection, URCNet can perform well on these

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two tasks at the same time. It will give more inspiration to build a general and unified deep network for remote sensing image processing tasks.

2) This article proposes various deep collaborative learning methods for URCNet optimization, which enforce the URCNet to support the multitask based on various application scenarios, including multitask based on the same scene images and multitask based on different scene images, even multimodal images.
3) This article conducts extensive experiments and analysis to verify the effectiveness and robustness of the proposed URCNet on image registration and change detection tasks. We also show the advantages of URCNet over existing deep networks for change detection under limited labeled datasets. URCNet can prevent significant performance degradation when the number of labeled datasets dramatically decreases.

The rest of this article is organized as follows. Section II introduces the related works of image registration and change detection. Section III presents our proposed unified deep network. Section IV shows the experimental results and analysis. Section V presents a conclusion of this article.

II. Related Work

This part will briefly introduce remote sensing image registration methods and change detection methods.

A. Image Registration

Remote sensing image registration methods can be divided into intensity-based and feature-based methods [4], [16]. Intensity-based methods directly maximize the image similarity between images for pixel alignment [17], [18]. The widely used image information similarity measurements are MI [7], CC [8], and their variants. Researchers also perform similarity matching and compute phase correlation for image registration by Fourier transformation [19]. The application of intensity-based methods is limited because of their high computational complexity and sensitivity to noise.

The feature-based approaches extract discriminative features from images using traditional algorithms [20] or deep networks. Then, many matching point pairs can be achieved according to feature distances and used for transformation matrix estimation. The features mainly contain point features, line features, and edge features [21]. The point features extracted from the keypoints on the images, which are relatively stable and simple, have attracted much research attention. Traditional feature-based registration methods include SIFT [9], SURF [10], Affine-SIFT [13], SAR-SIFT [12], and ORB [20], [22]. They are more robust to the image changes than image intensity in remote sensing image registration but less invariant to image modality changes in multimodal image registration.

Deep learning methods have shown significant performances on various remote sensing image processing tasks, such as land cover classification [23], [24], hyperspectral image classification [25], [26], [27], change detection [28], anomaly detection [29], [30], and object detection [31]. Recently, deep learning methods also have adopted the deep convolution network with multiple nonlinear mapping layers for feature extraction in image matching or registration. Deep features have higher invariance for matching samples and higher discrimination for identifying nonmatching samples than handcrafted features. Deep feature learning methods gradually replaced the traditional feature extraction methods and have acquired good performances in remote sensing image registration. Due to the down-sampling caused by pooling operations in the deep convolutional network, the location accuracy (ACC) of the keypoint needs to be updated. Thus, existing image registration methods use the traditional methods for keypoint detection and employ the deep network for feature extraction from the image patches surrounding the detected keypoints, such as SIFT points [9], Harris corners [32], and ORB points [22].

Deep learning methods contain metric learning methods and descriptor learning methods. The metric learning method converts the feature matching problem into a binary classification problem, directly predicting the matching label of the input image patch-pairs, such as matching or nonmatching. The metric learning network usually comprises a feature extraction network and a similarity metric network. The binary cross-entropy loss can be used for the metric learning network optimization [33], [34], [35]. The descriptor learning method mainly maps the local image patch into a high-dimensional feature space and achieves a feature vector. The matching samples will be closer than the nonmatching samples in the feature space. The descriptor learning networks widely use contrastive loss and triplet loss for optimization [15], [36], [37]. Contrastive loss constrains the feature distance of the matching samples as far as small, while the feature distance of the nonmatching samples is larger than a distance threshold. Triplet loss pays more attention to the relative feature distance between matching and nonmatching samples, which constrains the feature distance of nonmatching samples is larger than that of the matching samples by a margin. Due to the fast matching process, the descriptor learning methods are widely used for remote sensing image registration. The representative descriptor learning network is HardNet [15].

Although deep learning methods have shown significant advantages over traditional methods in the remote sensing image registration task, few studies consider whether the extracted features for image registration can be used for other tasks, such as change detection.

B. Change Detection

Remote sensing image change detection aims to identify the change regions between the coregistered multitemporal images. It can be divided into pixel-level, object-level, and scene-level change detection tasks according to the fundamental processing unit [38]. Pixel-level change detection typically involves analyzing and discriminating the change information of each pixel in the image and producing a binary change map, where the changed pixels are denoted by one and unchanged pixels are denoted by zero. The object-level change detection determines whether an object has changed

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and outputs object-level predictions. For instance, they use masks or bounding boxes to mark the changed objects. Scene-level change detection task mainly detects whether a category of the corresponding scene has changed and determines what kind of change has occurred. This article mainly focuses on pixel-level change detection and recognizes the change information of each pixel.

Traditional pixel-level change detection methods are mainly classified into algebra-based and transform-based methods. Algebra-based methods primarily obtain the change map by simple algebraic operations, such as image differencing [39], image regression [40], etc. The core of the algebra-based approach is the selection of threshold values to determine whether a pixel has changed. Transform-based methods enhance the change information of the multitemporal images by transforming the remote sensing image to another space [41], [42]. The representative transform-based methods are principal component analysis (PCA) [42], tasseled cap transformation [43], etc. Traditional algorithms are challenging to deal with complex scene changes, such as the significant influences of light, noise, shadows, and image style changes.

Due to the powerful feature extraction capability, several deep learning methods have been proposed for change detection and have significantly outperformed traditional methods. The deep network first extracts high-level features from images and generates the change map by comparing the feature differences of multitemporal images or directly predicting the change probability [44].

There are three crucial points for deep learning-based change detection. For example: 1) how to extract the features from the input multitemporal images; 2) where to fuse and compare the information of input multitemporal images; and 3) how to generate the final change map.

In terms of feature extraction, convolutional neural networks (CNNs) are used for feature learning from input images. Pixel-level change detection is similar to semantic segmentation, where the input images first through an encoder for feature learning and input into a decoder for generating the change map. For example, the U-Net structure is widely used for change detection because its up-sampling, down-sampling, and fusion structure of low-level and high-level features can effectively capture rich context information, reduce the information loss, and maintain the details of images [45], [46], [47], [48]. Papadomanolaki et al. [48] combines the U-Net structure and recurrent network of LSTMs for change detection. To further enhance the feature extraction, some attention modules are introduced into deep networks [6], [49], [50]. DSAMNet [6] used the convolutional block attention module (CBAM) [51] for remote sensing change detection, which can dynamically select important features along channel and spatial dimensions. Chen and Shi [52] developed a spatial-temporal attention module (STANet) to capture the relationships between the multitemporal images and produce better feature representations for change detection. To capture long-range dependencies between pixels and enlarge respective fields, Chen et al. [50] proposed a bitemporal image transformer (BIT) to extract rich context features for improving change detection performances.

In terms of the feature fusion, Daudt et al. [47] proposed three network structures to fuse and compare the information of a pair of images based on fully convolutional (FC) network and U-Net architecture, such as FC-early fusion (FC-EF), FC-Siam-conc, and FC-Siam-diff. FC-EF concatenates two input images along the channel dimensional, then extracts the features from the concatenated images, and predicts their change map. FC-Siam-conc first adopts a Siamese architecture with two same feature extraction branches to extract the features from the input images, respectively, and then combines the extracted features through the concatenation operator for predicting the change map. FC-Siam-diff also first extracts the features from the input images through a Siamese architecture and then computes the feature difference between the two input images to generate the change map. It can be seen that FC-EF fuses input images at the pixel level, while FC-Siam-conc and FC-Siam-diff merge the information of input images at the feature level through the concatenation and difference ways, respectively. These three deep network structures can effectively fuse multitemporal image information and achieve better change detection performances.

Additionally, the classification and metric method can be applied for change map generation. The former directly predicts the change label of each pixel, such as changed or unchanged. The latter computes the feature similarity or feature differences of input multitemporal images, which compares their similarity with the predetermined threshold value and produces the final change map. Generally, if the feature difference (similarity) is larger (smaller) than the threshold, the corresponding pixel is changed; otherwise, the corresponding pixel is unchanged.

As can be observed, the complexity of deep networks gradually increases to improve change detection performances. Therefore, the deep learning methods for change detection need a lot of labeled training datasets and expensive computing resources for network optimization. They will have a high risk of overfitting under limited training datasets.

C. Correlations Between Image Registration and Change Detection

As previously discussed, existing methods usually design two task-specific deep networks to extract features for image registration and change detection, respectively. The task-specific deep networks for different networks will increase the computation cost and parameter storage cost in remote sensing image applications. Additionally, the optimization of task-specific deep networks usually relies on a lot of labeled task-specific training datasets. We find that deep learning-based methods for image registration and change detection tasks focus on extracting discriminative features, which have a similar learning mechanism. Meanwhile, the image registration task is the foundation of change detection, where the multitemporal images should be registered before change detection. However, there is little research on combining image registration and change detection tasks. Existing deep learning methods view image registration and change detection as two

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fully independent tasks. This article focuses on investigating a unified deep framework for image registration and change detection.

The main differences between our proposed method and existing methods are as follows. Existing deep learning methods mainly focus on improving performance on a single task of image registration or change detection task. The deep network merely deals with a single task. The proposed URCNet can perform well on image registration and change detection tasks at the same time. Additionally, existing deep learning methods mainly adopt the labeled target task datasets for network optimization. Their change detection performances will degrade as the number of labeled datasets decreases. URCNet can employ the learned feature representation ability from image matching learning to boost change detection performances, reducing the dependency on labeled change detection datasets.

TABLE I
NETWORK STRUCTURE DETAILS OF THE URCNET

Blocks	Layers	Output channels
	Encoder	
Conv Block1	Conv-BN-ReLU	32
Conv Block2	Conv-BN-ReLU	32
Conv Block3	Conv-BN-ReLU	64
Conv Block4	Conv-BN-ReLU	64
Conv Block5	Conv-BN-ReLU	128
Conv Block6	Conv-BN-ReLU	128
Conv Block7	Dropout-Conv-BN	128
	Decoder	
AMFF Block1	SE-Deconv-BN-ReLU-Conv-BN-ReLU	128
AMFF Block2	SE-Deconv-BN-ReLU-Conv-BN-ReLU	128
AMFF Block3	SE-Deconv-BN-ReLU-Conv-BN-ReLU	64
AMFF Block4	SE-Deconv-BN-ReLU-Conv-BN-ReLU	64
AMFF Block5	SE-Deconv-BN-ReLU-Conv-BN-ReLU	32
AMFF Block6	SE-Deconv-BN-ReLU-Conv-BN-ReLU	32
AMFF Block7	SE-Dropout-Deconv-Conv	2
	CBAM	2

III. METHOD

The framework of the proposed united deep learning network for image registration and change detection is shown in Fig. 3. URCNet mainly contains an encoder and a decoder, where the image registration and change detection tasks share the encoder for feature extraction. For the image registration task, the encoder can obtain the feature descriptors from image patches for feature matching and local correspondence building. After that, we can align the images according to the acquired matching point-pairs. For the change detection task, the encoder can learn feature maps from the multitemporal images. Then, the extracted features are input into the decoder to generate a change map for change detection. We propose different deep collaborative learning methods for URCNet optimization based on the matching loss and cross-entropy loss. Deep collaborative learning methods will encourage the URCNet to support the image registration and change detection tasks simultaneously under the same or multimodal image datasets. The network structure and optimization strategies of the URCNet are shown in detail below.

A. Network Structure

The framework of the URCNet is shown in Fig. 3. It has a feature extraction module (encoder) and a feature decoding module (decoder). Image registration and change detection tasks share the feature extraction module. The image registration branch directly employs the encoder to extract feature descriptors for matching and registration. The change detection branch uses the encoder to extract the feature map from multitemporal images and takes the decoder to generate their change map.

1) Image Registration Branch: Feature descriptors learning is crucial for feature-based image registration. This article adopts the deep convolutional network to extract the feature descriptor for matching and then aligns images according to the acquired matching point-pairs. Considering the strong feature representation capacity and the small number of parameters of HardNet, URCNet utilizes the HardNet as an encoder to extract features for multitasks. HardNet contains seven convolutional blocks, the first six convolutional block has a convolutional layer, batch normalization (BN), and a nonlinear mapping function, rectified linear unit (ReLU). The final convolution block of HardNet has a dropout layer, a convolutional layer, and a BN. The encoder details are shown in Table I.

URCNet uses the encoder to extract the feature descriptors from the local image patches for the image registration task. The size of image patches is 64×64. Suppose there are N_r pairs of local image patches $(p_i^a, p_i^b), i = 1, \ldots, N_r$. (p_i^a, p_i^b) denotes matching image patch, while $(p_i^a, p_j^b), i \neq j$ denotes nonmatching image patch. The learned feature descriptor can be represented as follows:

$$\begin{aligned} f_i^a &= E\left(p_i^a\right) \\ f_i^b &= E\left(p_i^b\right) \end{aligned} \tag{1}$$

where E represents feature extraction operator through the encoder, f_i^a and f_i^b are the learned feature descriptors from the image patches of p_i^a and p_i^b.

The extracted feature descriptors should be normalized before matching. This article adopts the L_2 normalization, and the length of feature descriptors is 128. According to the extracted feature descriptors, we can build local correspondences between images through feature matching. After that, we estimate the transformation matrix through random sample consensus (RANSAC) and align images. The details of deep feature descriptors-based image registration can refer to works of Quan et al. [53], [54].

2) Change Detection Branch: For the change detection task, HardNet extracts the multilevel feature maps from the input multitemporal images. The low-level features have rich spatial information, while high-level features have semantic information. Thus, to boost the change detection performances, this article designs attention multilevel feature fusion (AMFF) blocks in the decoder, which can effectively and adaptively fuse the low-level and high-level features based on the attention mechanism.

The AMFF blocks are shown in Fig. 3. Each AMFF block has a squeeze-and-excitation (SE) block [55], deconvolutional layer (Deconv), and convolutional layer (Conv). The first

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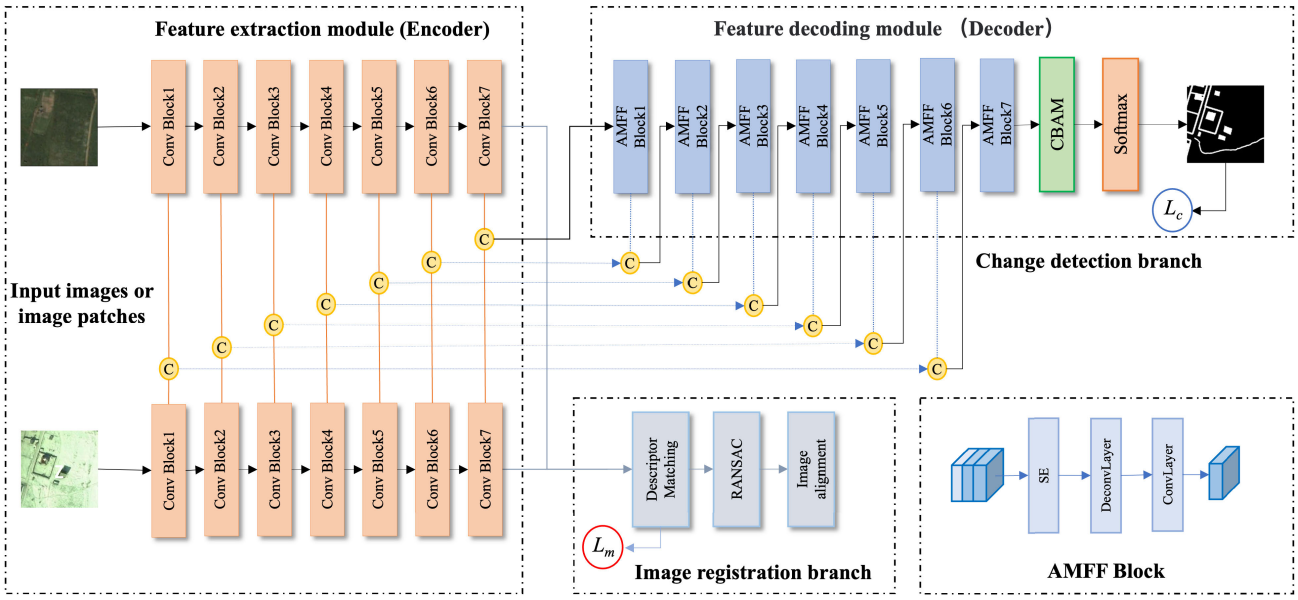


Fig. 3. Framework of the proposed URCNet. It can extract the feature descriptors for image registration and generate the change map for change detection. URCNet mainly contains a feature extraction module (encoder) and a decoder. The encoder has several convolutional blocks for feature learning. The decoder has several AMFF blocks and a CBAM to capture rich information for change detection.

AMFF block fuses high-level feature maps of the multitemporal images. Other AMFF blocks combine the feature maps of the multitemporal images and the output of the previous AMFF block. The feature maps of the multitemporal images at the same layer and the output of the previous AMFF block are fused by concatenation mode and then through the AMFF block. The carefully designed AMFF blocks will improve the change detection performances greatly. Finally, the fused multilevel features are input to a CBAM for predicting the change map. The decoder details are shown in Table I.

Suppose that the input multitemporal image pair is denoted by $\{(x_i^1, x_i^2), i = 1, \ldots, N_c\}$. The learned multilevel feature maps from images can be represented as follows:

$$\begin{aligned} f_{ij}^1 &= E\left(x_i^1\right) \\ f_{ij}^2 &= E\left(x_i^2\right) \end{aligned} \tag{2}$$

where f_{ij}^1 and f_{ij}^2 are the learned features of the jth layer of the ith image pairs of (x_i^1, x_i^2) through the encoder, $j = 1, \ldots, 7$.

The predicted change probability through the decoder based on multilevel features can be formulated as follows:

$$p_i = D\left(f_{ij}^1, f_{ij}^2\right) \tag{3}$$

where D represents decoder operator of the URCNet, p_i represents the predict change probability of ith image pairs. Finally, we can obtain the change label $\hat{y}_i$ according the corresponding change probability p_i.

B. Network Optimization

In feature-based image registration methods, extracting discriminative features and building local correspondences through image matching are very important. We usually adopt the triplet matching loss L_m for network optimization. L_m constrains the feature distance of the matching samples is smaller than that of the nonmatching samples by a margin. It can be formulated as follows:

$$L_m = \frac{1}{N_r} \sum_{i=1}^{N_r} \left[m + d\left(f_i^a, f_i^b\right) - d\left(f_i^a, f_j^b\right)\right]_+ \tag{4}$$

where $d(f_i^a, f_i^b)$ represents the Euclidean distance between the matching descriptors of (f_i^a, f_i^b), $d(f_i^a, f_j^b)$ represents the Euclidean distance between the nonmatching descriptors of $(f_i^a, f_j^b), i \neq j$, m is the distance margin, it is usually set to 1, and $[x]_+ = \max(0, x)$.

Change detection task can be viewed as the classification problem. Thus, we use cross-entropy loss L_c for deep change detection network training. L_c enforces the predicted change map as consistent with the ground truth. It can be represented as follows:

$$L_c = -\frac{1}{N_c} \sum_{i=1}^{N_c} \sum_{j=1}^{H \times W} y_{ij} \log\left(p_{ij}\right) \tag{5}$$

where y_{ij} represents the ground truth of the jth pixel in ith multitemporal image pairs, $j = 1, \ldots, H \times W$, H and W are the height and width of images, $y_{ij} = 1$ represents that the corresponding pixel is changed, and $y_{ij} = 0$ represents that the corresponding pixel is unchanged.

In this article, we expect the URCNet could support multitasks, image registration and change detection. It is very important to design an effective optimization strategy for URCNet. Therefore, this article proposes three different deep collaborative learning methods for network optimization, such as frozen learning, fine-tuning learning, and continuous learning, as shown in Fig. 4.

1) Frozen Learning: The extracted features for image registration and change detection tasks have high similarity. Thus, the learned features for image registration can be shared for change detection. As shown in Fig. 4 (left), the frozen learning method first trains the feature extraction module

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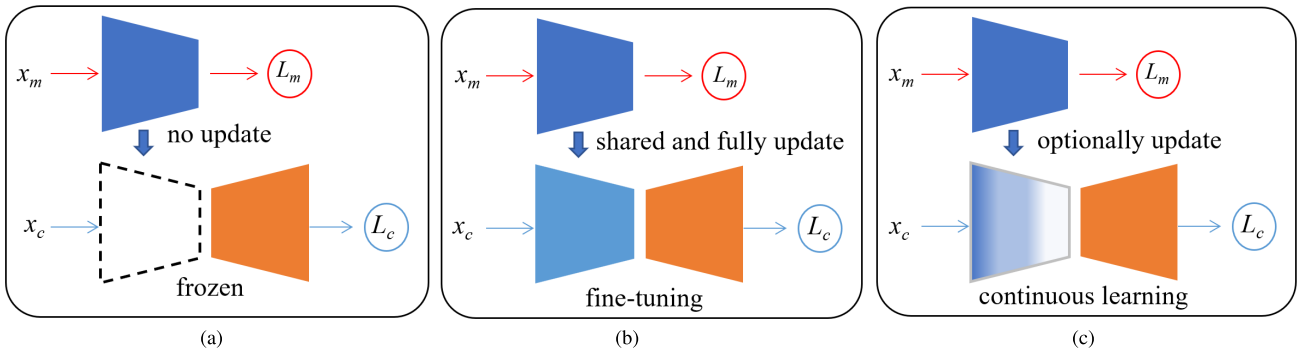


Fig. 4. Diagram of three deep collaborative learning strategies for URCNet optimization. (a) Frozen learning. (b) Fine-tuning learning. (c) Continuous learning.

for image matching based on L_m, then freezes the learned feature extraction module, and merely trains the decoder of the URCNet for the change detection based on L_c. The frozen learning method can flexibly choose the number of feature extraction layers to freeze. This article freezes the whole feature extraction module considering the number of network parameters. The frozen method can effectively prevent image matching and registration performance degradation. Meanwhile, it releases the network parameters in the decoder to adapt to the change detection task. Thus, the frozen learning method can encourage the URCNet to conduct image registration and change detection tasks effectively.

2) Fine-Tune Learning: In fine-tune learning method, the trained feature extraction module based on the matching loss can be viewed as a pretraining network for change detection. As shown in Fig. 4 (middle), the fine-tune learning method transfers the trained feature extraction module and fine-tune whole network parameters (encoder and decoder) for change detection. After the network training for change detection, the feature extraction module still can be applied for the image registration task.

3) Continuous Learning: The continuous learning method also can be used for multitask learning, which first trains the deep network for image registration, then optimizes the deep network for change detection, and meanwhile prevents the deep network occurs catastrophic forgetting in the previous task. As shown in Fig. 4 (right), the continuous learning method selects some parameters of the features extraction module for updating. Specifically, this article adopts the elastic weight consolidation (EWC) [56] for URCNet optimization. It optimizes the feature extraction module through the matching loss L_m and acquires the optimal parameters θ^*. Second, it computes the Fisher information matrix F of network parameters. Third, the continuous learning method adopts the EWC loss L_{ewc} and the cross-entropy for URCNet training, where the important parameters for matching and registration are updated less in the change detection task.

The continuous learning loss L for change detection can be represented as follows:

$$\begin{aligned} L &= L_c + L_{\text{ewc}} \\ L_{\text{ewc}} &= \sum_i \frac{\lambda}{2} F_i \left(\theta_i - \theta_i^*\right)^2 \end{aligned} \tag{6}$$

where θ is the parameter of the feature extraction module, and θ^* is the optimal parameter of the feature extraction module trained by the matching loss. λ is a regularization parameter of EWC. When the regularization parameter λ is larger, URCNet tends to maintain image registration performance with a smaller performance degeneration but will limit the change detection performance. When the regularization parameter λ is smaller, URCNet will release more parameters for change detection optimization, leading to better change detection performance but worse image registration performance.

As previously introduced, the frozen learning method merely trains the decoder parameters for change detection. The fine-tuning learning method should update the whole network parameters for change detection. The continuous learning method needs to optimize the decoder and the parameters in the encoder with lower importance for image registration in the subsequent task learning (change detection). The frozen learning method can keep the feature extraction capability and acquire high image registration performance. Meanwhile, a part of the network parameters in URCNet (decoder) can be optimized for the change detection task. In the processing of backpropagation, the computed gradient of the shallow layers is smaller than that of the deep layers. The weight parameters in the shallow module will have a relatively minor update. Additionally, due to the features of the deep network for the two tasks being similar, the parameter update of the feature extractor module is relatively small during fine-tuning. EWC restricts some important weights for image matching and registration with minor changes. Thus, fine-tune learning and continuous learning methods can support image registration and change detection tasks at the same time, and these two methods will acquire similar results. It should be noted that the iterative training method is unsuitable for URCNet optimization. The inputs of the two remote sensing image processing tasks are different. The image registration method aims to build local correspondences between images, which adopts the deep network to extract the features from input image patches. The input of the change detection network is multitemporal images. It is challenging to optimize the URCNet through the iterative training method.

From the discussion above, the proposed deep collaborative learning methods, frozen learning method, fine-tuning learning method, and continuous learning method, can effectively

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Fig. 5. Examples of SYSU dataset. The first two columns show multitemporal images, and the third column displays the ground truth of changes.



Fig. 6. Examples of CDD dataset. The first two columns show multitemporal images, and the third column displays the ground truth of changes.

enforce the URCNet to perform image registration and change detection tasks.

IV. Experiments

This part mainly tests the effectiveness of the proposed URCNet for image registration and change detection tasks. We present the influence of different deep collaborative learning methods on multitasks, the influence of the frozen layers in the encoder, and SE blocks in the AMFF module for change detection. This article compares the quantitative and visualization results of the URCNet with other deep networks. Meanwhile, we analyze the complexity of deep networks and the performances of change detection under different numbers of labeled training datasets.

A. Data Introduction

The experiments are conducted on three public change detection datasets, SYSU, CDD, and LEVIR, and a public multimodal remote sensing image dataset for image matching and registration, SEN1-2.

SYSU [6] contains 20 000 pairs of images with a resolution of 0.5 m taken from Hong Kong in 2007 and 2014. The image size is 256 × 256. SYSU dataset mainly contains the change information caused by the urban buildings, suburban dilation,



Fig. 7. Examples of LEVIR dataset. The first two columns show multitemporal images, and the third column displays the ground truth of changes.

vegetation changes, road expansion, and sea construction. The numbers of image pairs for training, testing, and validation are 12 000, 4000, and 4000, respectively. Some multitemporal images and corresponding ground truth of changes are shown in Fig. 5.

CDD [57] contains 16 000 pairs of images with a resolution of 0.03–1 m. The image size is 256 × 256. The main change information in the CDD dataset is car, tank, land, building, and road changes. The numbers of image pairs for training, testing, and validation are 10 000, 3000, and 3000, respectively. The multitemporal remote sensing images and corresponding ground truth of changes are shown in Fig. 6.

LEVIR [52] contains 637 pairs of images with a resolution of 0.3 m. The number of image pairs for training, testing, and validation is 445, 128 and 64, respectively. The image size is 1024 × 1024. The LEVIR dataset mainly focuses on the building changes. This article cropped images with a size of 256 × 256 from the LEVIR dataset, as shown in Fig. 7. The number of cropped images in the training set, test set, and validation set is 7120, 2048, and 1024, respectively.

SEN1-2 [58] consists of many OPT and SAR image pairs. The image size is 256 × 256, and the image resolution is 10 m. Local image matching is crucial for image registration, which will directly reflect the performance of deep networks for image registration. Thus, we test the local image matching results to present the registration performances. We selected 15 866 image pairs from the SEN1-2 dataset for image matching and registration. These images are divided into training and test sets according to 7:3. Some of the multimodal images from the SEN1-2 dataset are shown in Fig. 8, where the OPT and SAR images have significant differences caused by different imaging mechanisms. To optimize the deep network for image matching and registration, this article clips a lot of local image patches surrounding the SIFT keypoints. The local image patch size is 64 × 64. There are about 583 099 multimodal image patch-pairs for training and 248 010 for image matching evaluation.

In experiments, we present the change detection performances when the matching learning and change detection have different and the same scene training images. Thus,

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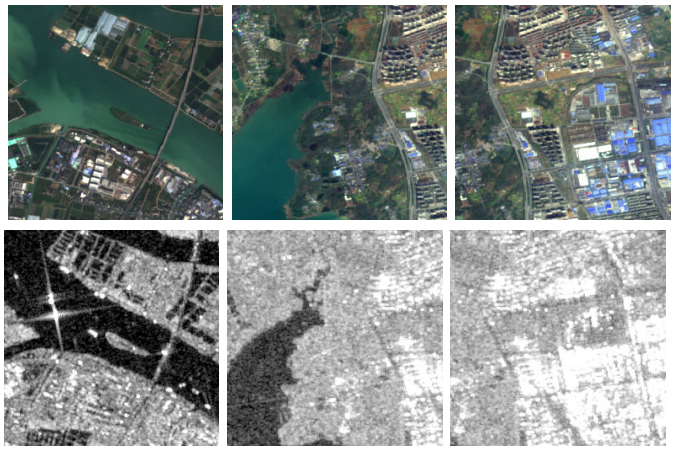


Fig. 8. Examples of SEN1-2 dataset. The first row shows the OPT images and the second row presents the corresponding SAR images.

we also generate local image patch-pairs from SYSU, CDD, and LEVIR datasets. The number of image patch-pairs for image matching learning of SYSU, CDD, and LEVIR is 670 324, 403 653, and 534 134, respectively.

B. Implement Details

All experiments were implemented on RTX 3090. The image registration used the Adam optimizer for network training with an initial learning rate of 1, $\beta 1 = 0.5$, $\beta 2 = 0.9$, and batch size of 500. The margin m in triplet matching loss L_m was 1. In continuous learning, λ is set to $1e^{20}$ to promise the performances of image matching and registration. The change detection adopted stochastic gradient descent (SGD) for network optimization with a weight decay of $1e^{-4}$, a momentum parameter of 0.9, and a batch size of 8.

C. Evaluation Metric

For feature-based image registration methods, local image matching performances can directly reflect the image registration results. This article uses FPR95 and matching ACC to evaluate the local image patch matching performance. FPR95 means the false positive rate when the true positive rate is 95%. The higher the ACC is, the better the matching performance is. The smaller FPR95 means a better matching performance.

Additionally, this article uses F1-score (F1), precision (P), recall (R), and intersection over union (IoU) to reflect the change detection performances. The higher these metrics are, the better the change detection performances are. These evaluation metrics are represented as follows:

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{7}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{8}$$

$$\mathrm{F1} = \frac{2PR}{P + R} \tag{9}$$

$$\mathrm{IoU} = \frac{\hat{y} \cap y}{\hat{y} \cup y} \tag{10}$$

where TP, FP, TN, and FN represent true positives, false positives, true negatives, and false negatives, respectively.

TABLE II

MATCHING AND CHANGE DETECTION RESULTS OF URCNET ARE BASED ON DIFFERENT OPTIMIZATION STRATEGIES. IT USES THE SEN1-2 DATASET FOR MATCHING AND SYSU DATASET FOR CHANGE DETECTION. HARDNET AND URCNET-O ARE TASK-SPECIFIC DEEP NETWORKS FOR MATCHING AND CHANGE DETECTION, RESPECTIVELY

Strategies	F1(%)	P(%)	R(%)	IoU(%)	FPR95 (%)	ACC(%)
HardNet	-	-	-	-	32.45	85.02
URCNet-O	79.33	77.50	81.26	65.75	-	-
Frozen	77.93	76.42	79.50	63.84	32.45	85.02
Continuous	77.50	79.03	76.02	63.26	80.40	65.26
Fine-tune	77.72	77.33	78.11	63.56	79.35	65.82

TABLE III

MATCHING AND CHANGE DETECTION RESULTS OF URCNET ARE BASED ON DIFFERENT OPTIMIZATION STRATEGIES. IT USES THE SEN1-2 DATASET FOR MATCHING AND CDD DATASET FOR CHANGE DETECTION

Strategies	F1(%)	P(%)	R(%)	IoU(%)	FPR95 (%)	ACC(%)
HardNet	-	-	-	-	32.45	85.02
URCNet-O	96.94	96.48	97.42	94.07	-	-
Frozen	94.17	93.12	95.25	88.98	32.45	85.02
Continuous	93.12	91.61	94.68	87.13	81.53	65.86
Fine-tune	93.28	92.50	94.08	87.41	80.89	65.71

TABLE IV

MATCHING AND CHANGE DETECTION RESULTS OF URCNET ARE BASED ON DIFFERENT OPTIMIZATION STRATEGIES. IT USES THE SEN1-2 DATASET FOR MATCHING AND LEVIR DATASET FOR CHANGE DETECTION

Strategies	F1(%)	P(%)	R(%)	IoU(%)	FPR95(%)	ACC(%)
HardNet	-	-	-	-	32.45	85.02
URCNet-O	88.93	87.21	90.71	80.06	-	-
Frozen	86.55	85.93	87.17	76.28	32.45	85.02
Continuous	86.68	84.66	88.80	76.49	82.46	64.94
Fine-tune	87.01	84.32	89.88	77.01	82.47	64.93

D. Effectiveness of Optimization Strategies

This article proposes three deep collaborative optimization strategies for URCNet training, such as frozen learning, fine-tuning, and continuous learning methods. To verify their effectiveness, we test matching results on SEN1-2 and change detection performances on SYSU, CDD, and LEVIR datasets based on different optimization strategies. It should be noted that the image matching dataset SEN1-2 contains multimodal images (OPT and SAR), while the change detection dataset only contains OPT images. URCNet should not only adapt to multitasks but also needs to be applied to different datasets and different image modalities, which are very challenging for deep networks. The experimental results are shown in Tables II–IV.

First, existing deep learning methods usually design two fully independent networks for image registration and change detection tasks, respectively. Their task-specific trained deep networks (HardNet and URCNet-O) can hardly be applied to another task. URCNet-O represents the deep network only trained from scratch for the single change detection task. However, the proposed URCNet can simultaneously be used for image registration and change detection. Second, the

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TABLE V

CHANGE DETECTION RESULTS OF URCNET ARE OPTIMIZED BY THE FROZEN METHOD BASED ON DIFFERENT FROZEN LAYERS. THE FEATURE EXTRACTION MODULE IS FIRST TRAINED ON THE SEN1-2 DATASET. THEN, WE OPTIMIZE THIS DEEP NETWORK FOR CHANGE DETECTION

Strategies	F1(%)	P(%)	R(%)	IoU(%)
SYSU dataset				
URCNet-O	79.33	77.50	81.26	65.75
Frozen-5	78.11	78.60	77.63	64.08
Frozen-6	77.91	77.50	79.56	63.82
Frozen-7	77.93	76.42	79.50	63.84
CDD dataset				
URCNet-O	96.94	96.48	97.42	94.07
Frozen-5	96.11	95.44	96.79	92.51
Frozen-6	94.23	93.63	94.83	89.09
Frozen-7	94.17	93.12	95.25	88.98
LEVIR dataset				
URCNet-O	88.93	87.21	90.71	80.06
Frozen-5	87.87	84.69	91.29	78.36
Frozen-6	87.26	84.66	90.02	77.39
Frozen-7	86.55	85.93	87.17	76.28

TABLE VI

EXPERIMENTAL RESULTS OF DEEP NETWORK WITH (W) OR WITHOUT (W/O) SE BLOCKS ON CHANGE DETECTION DATASETS

	F1(%)	P(%)	R(%)	IoU(%)
SYSU dataset				
w/o SE	78.37	**78.41**	78.33	64.43
w/ SE	**79.33**↑0.96	77.50 ↓0.91	**81.26**↑2.93	**65.75**↑1.32
CDD dataset				
w/o SE	96.65	96.45	96.86	93.52
w/ SE	**96.94**↑0.29	**96.48**↑0.03	**97.42**↑0.56	**94.07**↑0.55
LEVIR dataset				
w/o SE	86.88	86.77	86.99	76.80
w/ SE	**88.93**↑2.05	**87.21** ↑0.44	**90.71**↑3.72	**80.06** ↑3.26

frozen learning method can keep the matching performances without any degeneration. It means that URCNet acquires the same matching performances as the HardNet. The fine-tuning and continuous learning methods will decrease the matching results. Third, the URCNet with three optimization methods achieves comparable change detection performances with the change detection task-specific network, URCNet-O. There is a slight performance degeneration on change detection due to ensuring the matching performances. However, the significant advantage of the proposed URCNet is that it can be adapted to different tasks, different datasets, and even different image modalities. This article uses the frozen learning method in the other experiments.

E. Influence of Frozen Layers

The frozen learning method is flexible and can select different frozen layers in URCNet. This part mainly presents the influence of the frozen layers on change detection. As discussed in the work of Yosinski et al. [59], we froze the first several successive layers of the encoder to keep the coadapted features. For example, Frozen-"*n*" means freezing the first "*n*" convolutional blocks of the feature extraction modules trained by the matching loss, then optimizing the remaining network parameters through the change detection learning. "Frozen-7" represents that the whole feature extraction module is frozen. As shown in Table V, the deep network with fewer frozen layers tends to acquire better change detection performances. The main reason is that the URCNet with small frozen layers will have more adjustable parameters for change detection. However, the URCNet with fewer shared feature extraction layers will bring more network parameters for image registration and change detection tasks. Considering maximizing shared network parameters and minimizing whole network parameters, we froze the total feature extraction module in URCNet.

TABLE VII

CHANGE DETECTION RESULTS OF DIFFERENT DEEP LEARNING METHODS ON SYSU DATASET

Model	F1(%)	P(%)	R(%)	IoU(%)
FC-EF [47]	75.07	74.32	75.84	60.09
FC-Siam-diff [47]	72.57	**89.13**	61.21	56.96
FC-Siam-conc [47]	76.35	82.54	71.03	61.75
BiDateNet [48]	76.94	81.84	72.60	62.52
STANet [52]	77.37	70.76	**85.33**	63.09
DSAMNet [6]	78.18	74.81	81.86	64.18
URCNet-O	**79.33**	77.50	81.26	**65.75**
URCNet-SEN	77.93	76.42	79.50	63.84
URCNet-SYSU	78.32	77.87	78.78	64.37

F. Influence of SE Block

URCNet proposes an AMFF module to fuse the multilevel feature. It adopts the SE blocks to capture the relationships between feature channels and adaptively focus on important information to improve the feature representations. This part mainly verifies the effectiveness of the SE block on the change detection task. The change detection results with and without SE blocks are shown in Table VI. It can be seen that the introduced SE blocks significantly improve the change detection results. On the SYSU dataset, SE blocks improve F1 by 0.96%, increase R by 2.93%, and increase IoU by 1.32%. On the CDD dataset, SE blocks improve F1 by 0.29%, increase R by 0.56%, and improve IoU by 0.55%. On the LEVIR dataset, SE blocks improve F1 by 2.05%, increase R by 3.72%, and improve IoU by 3.26%. These experimental results present the significant advantages of the proposed AMFF module for change detection. Additionally, the SE blocks do not introduce many network parameters. Thus, we insert the SE block in URCNet for multilevel feature fusion.

G. Comparative Experimental Results on Change Detection

This part compares the proposed URCNet with some deep learning methods for change detection. The image registration and change detection tasks will deal with the same or different remote sensing images in particular. Thus, we test the performances of the URCNet for multitasks based on the same dataset and different scene datasets. In the image matching learning stage, the URCNet that is trained on SEN1-2, SYSU, CDD, and LEVIR datasets are denoted as URCNet-SEN, URCNet-SYSU, URCNet-CDD, and

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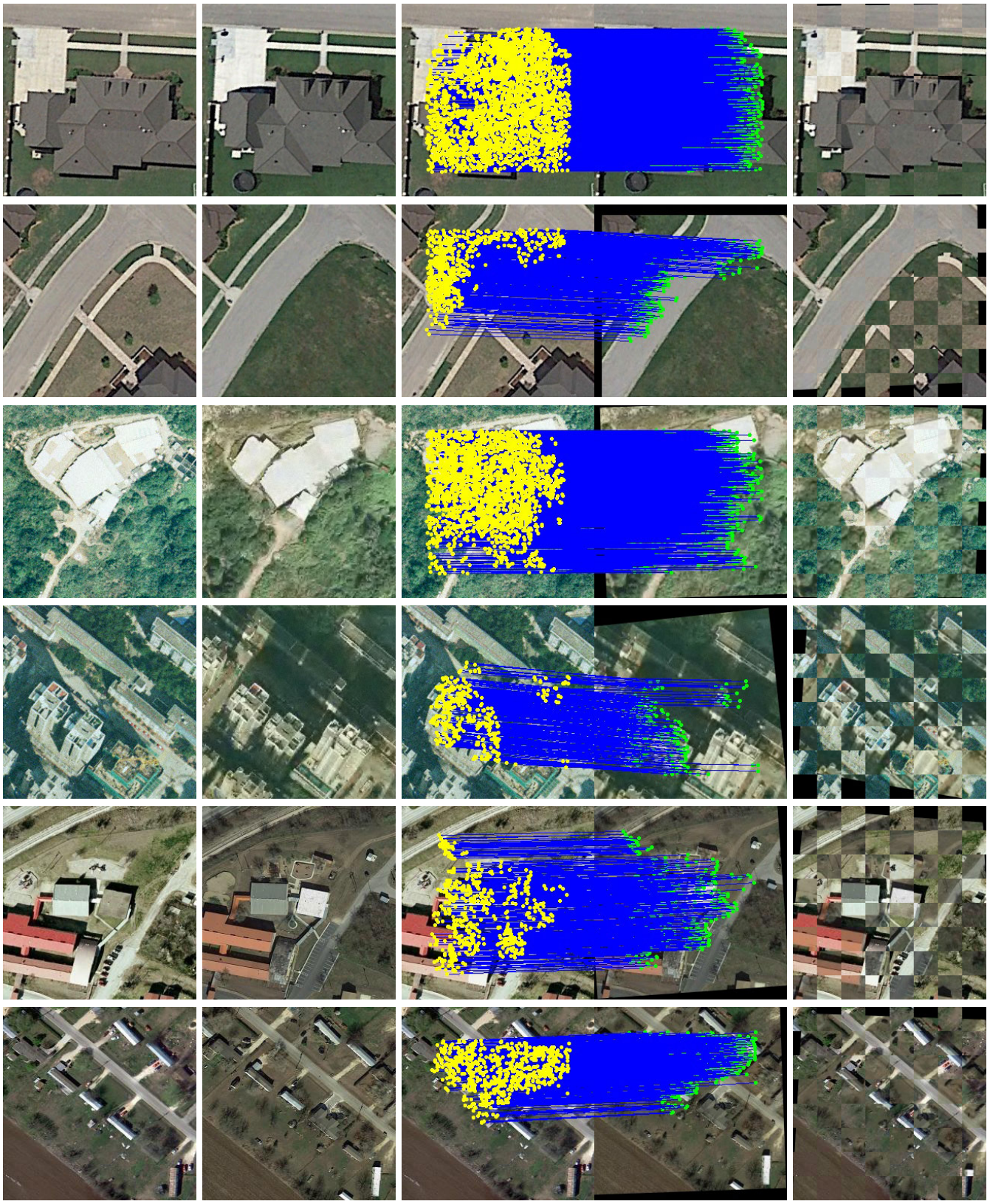
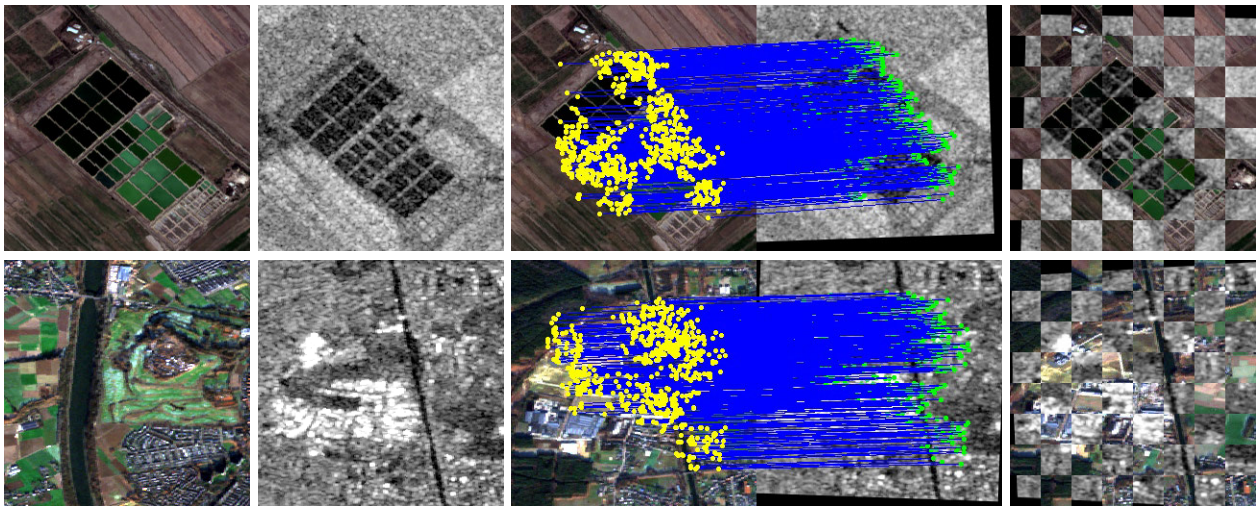


Fig. 9. URCNet performed the image registration results on OPT remote sensing image pairs of SYSU, CDD, and LEVIR datasets. The first and second columns are multitemporal images. We conduct random translation or rotation transformations on the aligned multitemporal images and then test the registration performances. The third column is the matching point-pairs images, and the fourth column is the checkboard images. (a) and (b) Input image pairs. (c) Matching point-pairs images. (d) Checkboard images.

URCNet-LEVIR, respectively. Experimental results are shown in Tables VII–IX.

Tables VII and VIII show that the proposed deep network acquires the best performances on SYSU and CDD datasets

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(a) (b) (c) (d)

Fig. 10. URCNet performed the image registration results on multimodal remote sensing image pairs of SEN1-2 dataset. The first and second columns are multimodal images, OPT and SAR images. We conduct random translation and slight rotation transformations on the aligned multimodal images and then test the registration performances. The third is the matching point-pairs images, and the fourth is the checkboard images. (a) OPT images. (b) SAR images. (c) Matching point-pairs images. (d) Checkboard images.

TABLE VIII

CHANGE DETECTION RESULTS OF DIFFERENT DEEP LEARNING METHODS ON CDD DATASET. ‡ MEANS MORE TRAINING ITERATIONS

Model	F1(%)	P(%)	R(%)	IoU(%)
FC-EF [47]	78.65	90.41	69.60	64.81
FC-Siam-diff [47]	82.93	94.76	73.71	70.84
FC-Siam-conc [47]	85.90	92.15	80.44	75.28
BiDateNet [48]	90.01	95.98	84.74	81.83
STANet [52]	91.44	89.28	93.71	84.23
DSAMNet [6]	93.69	94.54	92.77	88.13
URCNet-O	**96.94**	**96.48**	**97.42**	**94.07**
URCNet-SEN	94.17	93.12	95.25	88.98
URCNet-CDD	94.89	94.30	95.49	90.28
URCNet-O ‡	**98.09**	**97.85**	**98.34**	**96.25**
URCNet-SEN ‡	96.05	95.44	96.66	92.39
URCNet-CDD ‡	96.40	95.85	96.96	93.05

TABLE IX

CHANGE DETECTION RESULTS OF DIFFERENT DEEP LEARNING METHODS ON LEVIR DATASET. ‡ MEANS MORE TRAINING ITERATIONS

Model	F1(%)	P(%)	R(%)	IoU(%)
FC-EF [47]	89.23	**91.31**	87.38	**82.01**
FC-Siam-diff [47]	87.47	83.17	**92.22**	77.72
FC-Siam-conc [47]	85.56	79.99	92.02	74.81
BiDateNet [48]	89.32	86.68	92.12	80.70
STANet [52]	87.30	86.70	87.90	77.50
DSAMNet [6]	84.36	86.34	82.46	72.94
URCNet-O	88.93	87.21	90.71	80.06
URCNet-SEN	86.55	85.93	87.17	76.28
URCNet-LEVIR	87.50	84.84	90.32	77.77
URCNet-O ‡	**89.92**	88.51	91.38	81.69
URCNet-SEN ‡	87.24	84.51	90.15	77.37
URCNet-LEVIR ‡	87.45	84.32	90.83	77.70

in terms of the single change detection task. From Table IX, we can find that our model achieves similar results with the best results on the LEVIR dataset. These results demonstrated the effectiveness of the proposed deep network structure. The change detection performances of the proposed deep model will be further improved when there are more training iterations for CDD and LEVIR datasets. The reason may be the smaller number of training image pairs for change detection on CDD and LEVIR datasets than that of the SYSU dataset. Additionally, the change label of the CDD dataset is more refined, and the LEVIR dataset mainly focuses on building changes. Thus, the change detection task on the CDD and LEVIR datasets is more difficult than on the SYSU dataset.

For multitasks, the change detection results of URCNet have a slight decrease compared with the URCNet-O. The main reason is that the URCNet should consider the performances of image registration and change detection tasks. However, existing deep learning methods are merely suitable for the change detection task, which cannot be applied to image registration. With the increase of training iterations, the change detection performance gains of URCNet-SEN and URCNet-CDD are larger than URCNet-O on the CDD dataset, which will further narrow the performance differences between the URCNet and URCNet-O. Additionally, when the multitasks deal with different scene images and even multimodal images, the proposed URCNet still can keep high performances. If the training datasets for image registration and change detection are from the same scene, the URCNet will obtain better change detection performances. For example, URCNet-SYSU, URCNet-CDD, and URCNet-LEVIR have slightly better results than URCNet-SEN on the SYSU, CDD, and LEVIR datasets, respectively.

These experimental results show the strong robustness of the proposed URCNet, which not only deals with the multitasks based on the same scene images but also performs the multitasks based on different scene images and multimodal images. The main reasons are analyzed as follows. URCNet adopts multiple convolutional layers to extract high-level features from images. Additionally, URCNet shares the same parameters in the feature extraction module for multimodal images. It enforces the extracted features of the matching

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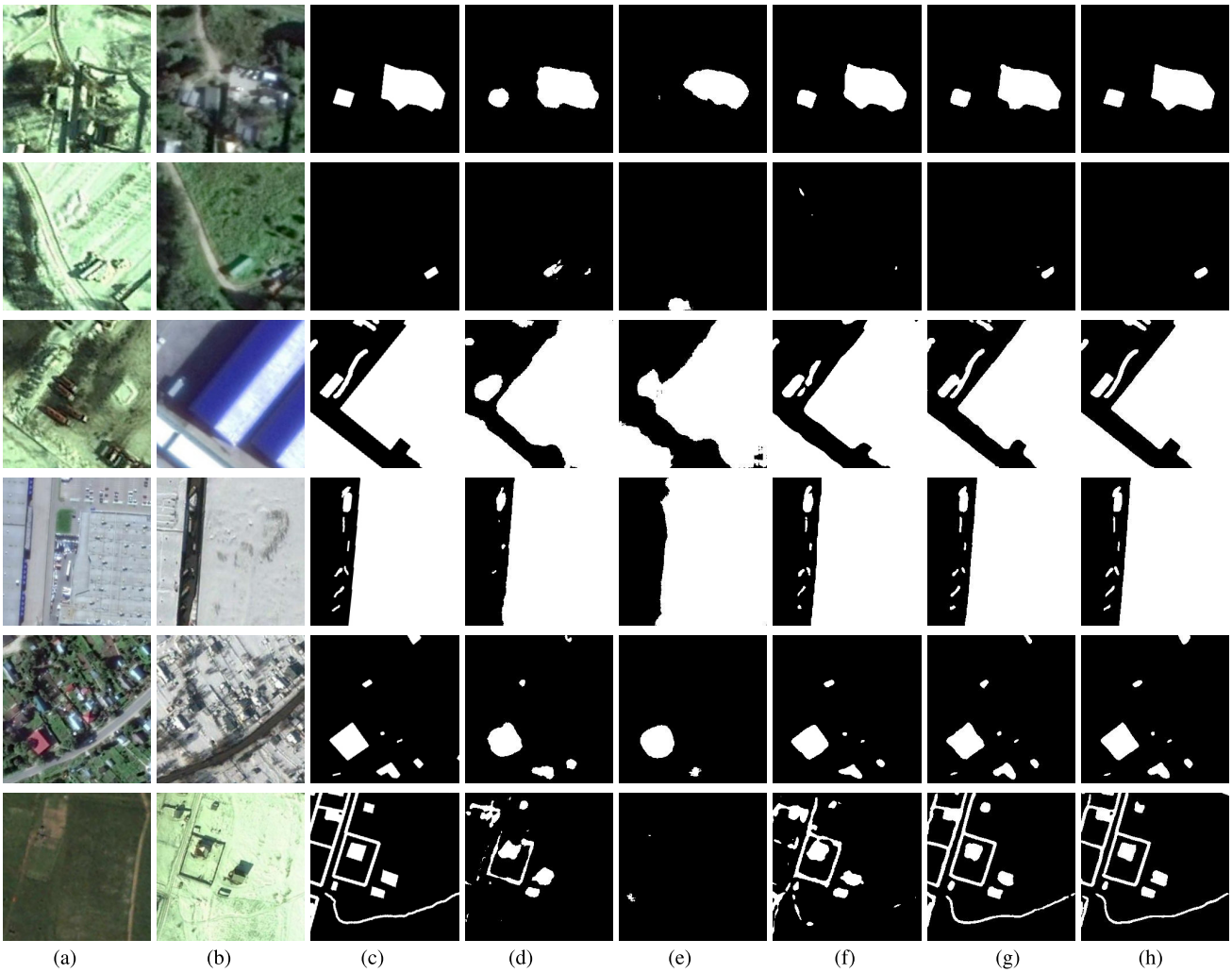


Fig. 11. Visual comparison results on the CDD dataset. (a) and (b) Input multitemporal images. (c) Corresponding ground truth. (d) Detection results of FC-Siam-conc. (e) Detection results of FC-Siam-diff. (f) Detection results of DSAMNet. (g) Detection results of URCNet. (h) Detection results of URCNet-O.

TABLE X
COMPARATIVE EXPERIMENTAL RESULTS ON PARAMETERS AND FLOPS

	Param(M)	FLOPs(G)
FC-Siam-conc	1.55	5.33
FC-EF	1.35	3.58
FC-Siam-diff	1.35	4.73
BiDateNet	13.40	45.80
DSAMNet	16.95	75.39
URCNet	4.55	11.16

multimodal images to be similar. In contrast, the extracted features of the nonmatching samples are dissimilar. The extracted high-level features are robust to the image modality changes. Thus, URCNet can be used for both multimodal and single-modal datasets.

H. Complexity of Networks

This part compares the complexity of the proposed URCNet with other deep learning networks. Table X shows the number of parameters and the floating point operations (FLOPs) of different change detection methods. The size of the input images is 256 × 256. Existing deep networks for change detection usually adopt the color images with three channels, while the URCNet uses the gray images with a single channel. As shown in Table X, although the number of FC-EF, FC-Siam-diff, and FC-Siam-conc model parameters and FLOPs are smaller than those of URCNet, their change detection performances are worse than those of URCNet. Additionally, URCNet acquires comparable and better performances with the more complex deep networks, BiDateNet and DSAMNet. The numbers of parameters of BiDateNet and DSAMNet are 13.4 and 16.95, while that of the URCNet is 4.55. The FLOPs of BiDateNet and DSAMNet are 45.8 and 75.39, while that of the URCNet is 11.16. URCNet has significant advantages over BiDateNet and DSAMNet in the network complexity. Moreover, URCNet can perform change detection and image registration tasks at the same time.

I. Visualization Results of URCNet

To demonstrate that URCNet can perform the image registration and change detection tasks well, we present the visualization results of the URCNet on these two tasks. First, Fig. 9 shows the image registration results on OPT remote sensing image pairs, which have slight translation or

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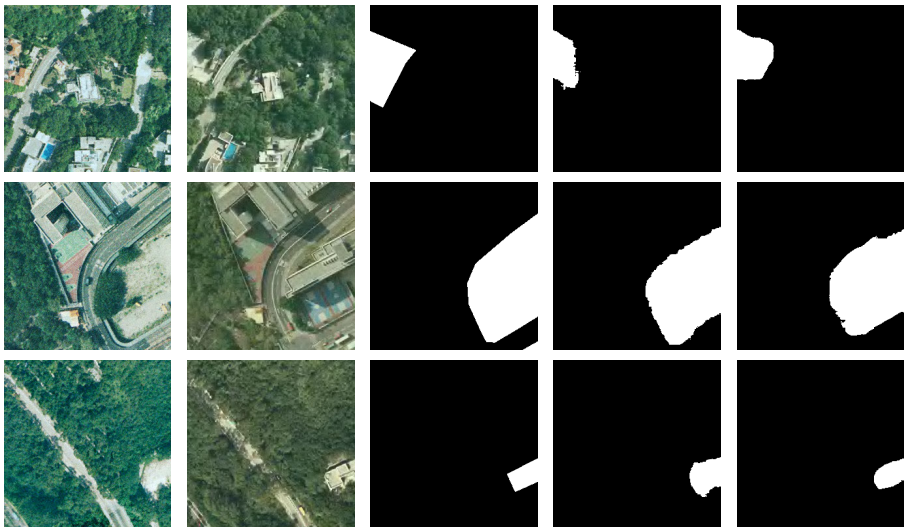


Fig. 12. Change detection results on the SYSU dataset. (a) and (b) Input multitemporal images. (c) Corresponding ground truth. (d) Detection results of URCNet. (e) Detection results of URCNet-O.

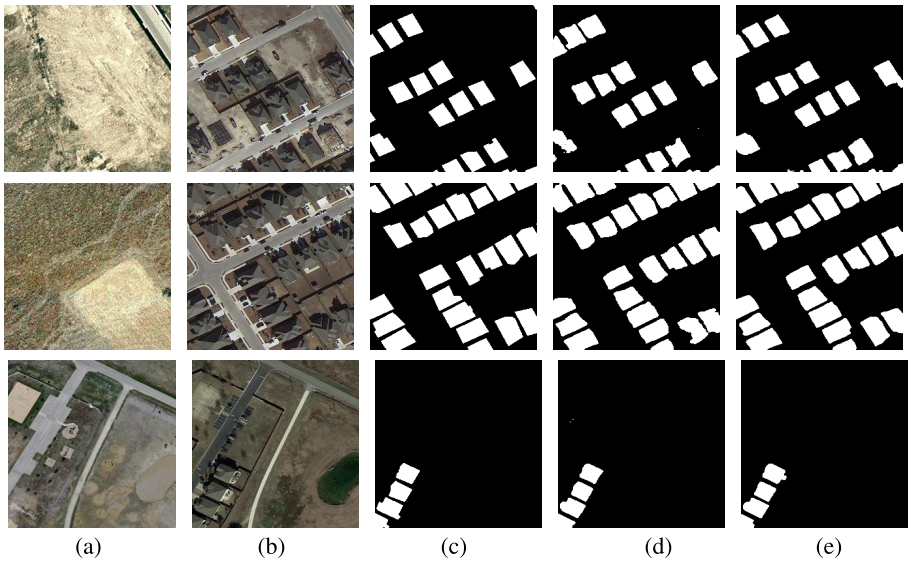


Fig. 13. Change detection results on the LEVIR dataset. (a) and (b) Input multitemporal images. (c) Corresponding ground truth. (d) Detection results of URCNet. (e) Detection results of URCNet-O.

rotation changes. From the matching point-pairs images and checkboard images, it can be seen that URCNet can capture many matching point-pairs and align images accurately. Fig. 10 presents the multimodal remote sensing image registration performances on OPT and SAR images. Although OPT and SAR images have significant appearance differences, URCNet still acquires many matching points and registers multimodal images.

From Fig. 11, we can see that FC-Siam-conc and FC-Siam-diff can detect obvious changes but will ignore many small changes. The generated change maps by URCNet are close to the corresponding ground truth, where the obvious and slight changes can be well detected. Figs. 12 and 13 show the change detection results on SYSU and LEVIR datasets. We can see that the detection results of URCNet are similar to the corresponding ground truth. These visualization results also demonstrated the effectiveness of the proposed URCNet on image registration and change detection tasks.

J. Influence of the Number of Training Dataset

It is well known that existing deep learning networks for change detection require large amounts of labeled training datasets for optimization. However, pixel-level labeling for change detection training is very expensive. As shown in Fig. 14, when the proportions of the training dataset for change detection decrease from 100% to 50%, 10%, and 5%, the change detection results (F1 and IoU) of FC-EF, FC-Siam-diff, FC-Siam-conc, and DSAMNet will drop on SYSU dataset. This decrease trend is more significant on the CDD dataset. The decline of the URCNet is slower than existing deep learning methods. We think that there are two main reasons as follows. On the one hand, URCNet has a smaller number of network parameters than other deep networks. Thus, URCNet can avoid falling into overfitting under limited labeled datasets. On the other hand, the frozen feature extraction module trained by matching learning will

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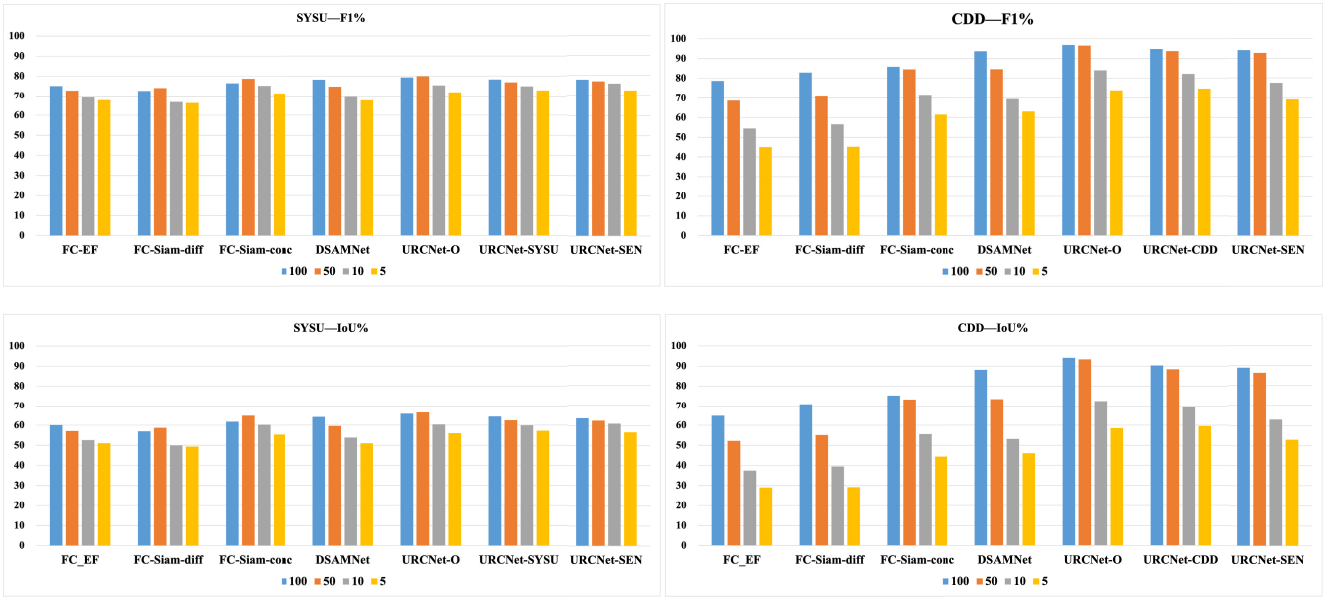


Fig. 14. F1% and IoU% results of the SYSU dataset- and CDD dataset-based different proportions of the training datasets, such as 100%, 50%, 10%, and 5%.

further enhance the network optimization for change detection. URCNet has stronger feature representation ability than the network that is trained from scratch. Therefore, our URCNet performs better than other deep learning methods based on the limited labeled change detection training datasets.

V. Conclusion

This article proposes a unified deep learning framework for image registration and change detection tasks, URCNet. Meanwhile, we propose three effective deep collaborative learning methods for URCNet optimization. The proposed URCNet shares the feature extraction module for multitasks, which can support image registration and change detection tasks at the same time. Additionally, this article designs the AMFF block to enhance the feature fusion and improve the change detection performances. Experimental results have shown the effectiveness of URCNet on multitasks, which can conduct image registration and change detection based on the same scene images, different scene images, and multimodal images. It can achieve comparable and better change detection performances with more complex deep networks. We also prove that URCNet has significant advantages over other deep networks in change detection under limited labeled training datasets.

Although the URCNet can perform change detection and image registration tasks simultaneously, the change detection results of URCNet are slightly worse than those of the task-specific deep network. The main reason is that URCNet needs to maintain high image registration performances in the network optimization process for change detection. It inevitably decreases the performance of change detection. Additionally, image registration learning can boost URCNet optimization under the limited labeled change detection datasets, while change detection learning does not promote network optimization for image registration. Thus, we will study a more effective optimization strategy for URCNet training, where these two tasks can promote each other. Additionally, we will study a more robust deep network to the various image degradation in future work.

References

[1] D. M. Tralli, R. G. Blom, V. Zlotnicki, A. Donnellan, and D. L. Evans, "Satellite remote sensing of earthquake, volcano, flood, landslide and coastal inundation hazards," *ISPRS J. Photogramm. Remote Sens.*, vol. 59, no. 4, pp. 185–198, Jun. 2005.

[2] C. J. Van Westen, "Remote sensing and GIS for natural hazards assessment and disaster risk management," *Treatise Geomorphol.*, vol. 3, pp. 259–298, Mar. 2013.

[3] B. Zitová and J. Flusser, "Image registration methods: A survey," *Image Vis. Comput.*, vol. 21, no. 11, pp. 977–1000, Oct. 2003.

[4] R. Feng, H. Shen, J. Bai, and X. Li, "Advances and opportunities in remote sensing image geometric registration: A systematic review of state-of-the-art approaches and future research directions," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 120–142, Dec. 2021.

[5] N. Bourdis, D. Marraud, and H. Sahbi, "Constrained optical flow for aerial image change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2011, pp. 4176–4179.

[6] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.

[7] J. P. W. Pluim, J. B. A. Maintz, and M. A. Viergever, "Mutual-information-based registration of medical images: A survey," *IEEE Trans. Med. Imag.*, vol. 22, no. 8, pp. 986–1004, Aug. 2003.

[8] Y. Wu, W. Ma, M. Gong, L. Su, and L. Jiao, "A novel point-matching algorithm based on fast sample consensus for image registration," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 43–47, Jan. 2015.

[9] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.

[10] H. Bay, A. Ess, T. Tuytelaars, and L. Van Gool, "Speeded-up robust features (SURF)," *Comput. Vis. Image Understand.*, vol. 110, no. 3, pp. 346–359, Jun. 2008.

[11] S. Suri, P. Schwind, J. Uhl, and P. Reinartz, "Modifications in the SIFT operator for effective SAR image matching," *Int. J. Image Data Fusion*, vol. 1, no. 3, pp. 243–256, Sep. 2010.

[12] F. Dellinger, J. Delon, Y. Gousseau, J. Michel, and F. Tupin, "SAR-SIFT: A SIFT-like algorithm for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 453–466, Jan. 2015.

[13] G. Yu and J.-M. Morel, "ASIFT: An algorithm for fully affine invariant comparison," *Image Process. Line*, vol. 1, pp. 11–38, Feb. 2011.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:22:46 UTC from IEEE Xplore. Restrictions apply.

[14] E. Simo-Serra, E. Trulls, L. Ferraz, I. Kokkinos, P. Fua, and F. Moreno-Noguer, "Discriminative learning of deep convolutional feature point descriptors," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 118–126.

[15] A. Mishchuk, D. Mishkin, F. Radenovic, and J. Matas, "Working hard to know your neighbor's margins: Local descriptor learning loss," 2017, *arXiv:1705.10872*.

[16] J. Ma, X. Jiang, A. Fan, J. Jiang, and J. Yan, "Image matching from handcrafted to deep features: A survey," *Int. J. Comput. Vis.*, vol. 129, no. 1, pp. 23–79, Jan. 2021.

[17] M. P. S. Tondewad and M. M. P. Dale, "Remote sensing image registration methodology: Review and discussion," *Proc. Comput. Sci.*, vol. 171, pp. 2390–2399, Jan. 2020.

[18] P. Bunting, R. Lucas, and F. Labrosse, "An area based technique for image-to-image registration of multi-modal remote sensing data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2008, p. 212.

[19] X. Tong et al., "Image registration with Fourier-based image correlation: A comprehensive review of developments and applications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 10, pp. 4062–4081, Oct. 2019.

[20] D. G. Lowe, "Object recognition from local scale-invariant features," in *Proc. 7th IEEE Int. Conf. Comput. Vis.*, Sep. 1999, pp. 1150–1157.

[21] M. Wyawahare, P. Patil, and H. Abhyankar, "Image registration techniques: An overview," *Int. J. Signal Process., Image Process. Pattern Recognit.*, vol. 2, no. 3, pp. 11–28, 2009.

[22] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 2564–2571.

[23] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[24] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Jun. 2023.

[25] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[26] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[27] T. Guo, R. Wang, F. Luo, X. Gong, L. Zhang, and X. Gao, "Dual-view spectral and global spatial feature fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, May 2023.

[28] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Jan. 2023.

[29] C. Li, B. Zhang, D. Hong, J. Yao, and J. Chanussot, "LRR-Net: An interpretable deep unfolding network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, May 2023.

[30] N. Huyan, X. Zhang, D. Quan, J. Chanussot, and L. Jiao, "Cluster-memory augmented deep autoencoder via optimal transportation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jun. 2022.

[31] S. Wang et al., "Select, purify, and exchange: A multisource unsupervised domain adaptation method for building extraction," *IEEE Trans. Neural Netw. Learn. Syst.*, Jul. 2023.

[32] C. G. Harris and M. Stephens, "A combined corner and edge detector," in *Proc. Alvey Vis. Conf.*, vol. 15, no. 50, 1988, p. 5244.

[33] C. A. Aguilera, F. J. Aguilera, A. D. Sappa, C. Aguilera, and R. Toledo, "Learning cross-spectral similarity measures with deep convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2016, pp. 267–275.

[34] S. Zagoruyko and N. Komodakis, "Learning to compare image patches via convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 4353–4361.

[35] X. Han, T. Leung, Y. Jia, R. Sukthankar, and A. C. Berg, "MatchNet: Unifying feature and metric learning for patch-based matching," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3279–3286.

[36] Y. Tian, X. Yu, B. Fan, F. Wu, H. Heijnen, and V. Balntas, "SOSNet: Second order similarity regularization for local descriptor learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11008–11017.

[37] Y. Tian, B. Fan, and F. Wu, "L2-Net: Deep learning of discriminative patch descriptor in Euclidean space," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6128–6136.

[38] L. Zhang, X. Hu, M. Zhang, Z. Shu, and H. Zhou, "Object-level change detection with a dual correlation attention-guided detector," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 147–160, Jul. 2021.

[39] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000.

[40] P. R. Coppin and M. E. Bauer, "Digital change detection in forest ecosystems with remote sensing imagery," *Remote Sens. Rev.*, vol. 13, nos. 3–4, pp. 207–234, Apr. 1996.

[41] Z. Liangpei and W. Chen, "Advance and future development of change detection for multi-temporal remote sensing imagery," *Acta Geodaetica et Cartographica Sinica*, vol. 46, no. 10, p. 1447, 2017.

[42] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[43] E. P. Crist, "A TM tasseled cap equivalent transformation for reflectance factor data," *Remote Sens. Environ.*, vol. 17, no. 3, pp. 301–306, Jun. 1985.

[44] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[45] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[46] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Munich, Germany. Cham, Switzerland: Springer, Oct. 2015, pp. 234–241.

[47] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[48] M. Papadomanolaki, S. Verma, M. Vakalopoulou, S. Gupta, and K. Karantzalos, "Detecting urban changes with recurrent neural networks from multitemporal Sentinel-2 data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 214–217.

[49] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.

[50] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jul. 2021.

[51] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.

[52] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[53] D. Quan et al., "Efficient and robust: A cross-modal registration deep wavelet learning method for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, May 2023.

[54] D. Quan et al., "Deep feature correlation learning for multi-modal remote sensing image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Aug. 2022.

[55] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[56] K. James et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Nat. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, Mar. 2017.

[57] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. XLII-2, pp. 565–571, May 2018.

[58] M. Schmitt, L. H. Hughes, C. Qiu, and X. X. Zhu, "SEN12MS—A curated dataset of georeferenced multi-spectral Sentinel-1/2 imagery for deep learning and data fusion," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. IV-2/W7, pp. 153–160, Sep. 2019.

[59] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How transferable are features in deep neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 27, 2014.

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